

Q1) The System Administrator has set the following rules for the password:

- ✓ The password is a string made of 10 characters.
- ✓ The available characters are Hexadecimal digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, and F.

Repetition is allowed unless otherwise stated

a) How many passwords with at least seven A's and at least one B, have no adjacent B's?

Explanation: Consider the following table that summarises all cases:

7A 1B 2 other char	8A 1B 1 other char	9A 1B 0 other char	7A 2B 1 other char	8A 2B 0 other char	7A 3B 0 other char
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All cases listed above are mutually exclusive, have no overlap, and can be added without overcounting.

Other characters (none of which are A/B) have 14 choices – let # be a place holder for it.

- **1B**
 - 7As and 1B and 2 other char

The number of different strings of AAAAA AAB## where each # has 14 options is $\frac{10!}{7!2!} \cdot (14)^2$.

- 8As and 1B and 1 other char

The number of different strings of AAAAA AAAB# where each # has 14 options is $\frac{10!}{8!} \cdot (14)$.

- 9As and 1B

The number of different strings of AAAAA AAAAB is $\frac{10!}{9!}$.

- **2Bs**
 - 7As and 2Bs and 1 other char

B's can be kept non-adjacent in a string AAAAA AA# by inserting them in any 2 of the available 9 positions “_”:

_A_A_A_A_A_A_A_#_

There are $\binom{9}{2}$ ways to insert B's and $\frac{8!}{7!} \cdot (14)$ different strings made of AAAAA AA#. Hence,

$\binom{9}{2} \cdot \frac{8!}{7!} \cdot (14)$ strings with no adjacent B's.

- 8As and 2Bs

$$\frac{a}{b} x^y \binom{n}{k}$$

B's can be kept non-adjacent in a string AAAAA AAA by inserting them in any 2 of the available 9 positions “_”:

_A_A_A_A_A_A_A_A_

There are $\binom{9}{2}$ ways to insert B's and therefore $\binom{9}{2}$ such strings.

- **3Bs**
 - 7As and 3Bs

B's can be kept non-adjacent in a string AAAAA AA by inserting them in any 2 of the available 8 positions “_”:

_A_A_A_A_A_A_A_

There are $\binom{8}{3}$ ways to insert B's and therefore $\binom{8}{3}$ such strings.

Answer: $\frac{10!}{7!2!} \cdot (14)^2 + \frac{10!}{8!} \cdot (14) + \frac{10!}{9!} + \binom{9}{2} \cdot \frac{8!}{7!} \cdot (14) + \binom{9}{2} + \binom{8}{3}$

b) How many passwords have exactly two distinct digits? AABBA AAAAB is an example of such string.

Explanation: Selecting two distinct digits, from 16 Hex digits can be done in $\binom{16}{2}$ different ways.

Suppose that we have selected A and B. There are 2^{10} distinct passwords made of A's and B's, some are:

AAAAA AAAAA
 AAAAA AAAAB
 AAAAA AAABA
 AAAAA AAABB
 ...
 BBBBB BBBA
 BBBBB BBBAB
 BBBBB BBBBA
 BBBBB BBBB

However, the first and the last passwords are made of A's only and B's only, and the remaining $2^{10} - 2$ passwords have two distinct characters.

Since there are $\binom{16}{2}$ different pairs, and for every pair, there are $2^{10} - 2$ different passwords, there is a total of $\binom{16}{2}(2^{10} - 2)$ different passwords that have exactly two distinct digits.

Answer: $\binom{16}{2}(2^{10} - 2)$

$$\frac{a}{b} \quad x^y \quad \binom{n}{k}$$

- c) How many passwords have six digits from the set 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 (repetition is allowed) and four digits from the set A, B, C, D, E, F (no repetition is allowed)?

Explanation: Let X be a placeholder for a digit from the set 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 and Y a placeholder for a digit from the set A, B, C, D, E, F. The placeholders can be shuffled in $\frac{10!}{6!4!}$ different ways, some examples:

XXXXX XYYYY
XXXXX YYYYYX
YXYXY YXXXX
...
YYYYX XXXXX

Consider any shuffle above:

- Six X placeholders are restricted to six digits from the set 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 (repetition is allowed). This can be done in 10^6 different ways.
- Four Y placeholders are restricted to four digits from the set A, B, C, D, E, F (no repetition is allowed). This can be done in $P(6,4) = \frac{6!}{2!}$ different ways.

Hence, for every shuffle of X's and Y's there are $(10^6) \frac{6!}{2!}$ different passwords. Since the number of shuffles is $\frac{10!}{6!4!}$, and the number of passwords is $(10^6) \frac{6!}{2!}$ per shuffle, the total is $\frac{10!}{6!4!} (10^6) \frac{6!}{2!}$

Answer: $\frac{10!}{6!4!} (10^6) \frac{6!}{2!}$

- d) How many passwords have sum of all digits equal to 18? Some examples of such passwords are 0A000 10007, 11226 21111 and 00200 0001F?

Explanation: Let $X_1, X_2, \dots, X_{10}$ represent 10 digits. Then, the number of ways we can have $X_1, X_2, \dots, X_{10}$ sum up to 18, where $X_1, X_2, \dots, X_{10} \geq 0$ is equal to the number of different solutions to the equation

$X_1 + X_2 + \dots + X_{10} = 18$, which is $\binom{10+18-1}{18} = \binom{27}{18}$. Some solutions are shown below:

X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}
0	10	0	0	0	0	1	0	0	7
1	1	2	2	3	3	1	1	1	3
0	0	2	0	0	0	0	0	1	15
2	2	0	0	0	0	0	0	0	14
18	0	0	0	0	0	0	0	0	0
17	0	0	0	0	0	0	0	0	1

$$\frac{a}{b} \quad x^y \quad \binom{n}{k}$$

0	0	0	0	16	0	0	0	2	0
0	16	1	0	0	0	1	0	0	0

However, $\binom{27}{18}$ also includes solutions where a bin/digit has values 18, 17 or 16 (in bold in the table above). In Hex, the highest digit is F=15 and the following 4 cases need to be taken away:

1. One 18, nine 0's
2. One 17, one 1, eight 0's
3. One 16, one 2, eight 0's
4. One 16, two 1's, seven 0's

Using Mississippi approach on each case and summing them up, gives:

$$10 + \frac{10!}{8!} + \frac{10!}{8!} + \frac{10!}{2!7!}$$

So, subtracting the non-existing solutions from the total gives then $\binom{27}{18} - \left(10 + \frac{10!}{8!} + \frac{10!}{8!} + \frac{10!}{2!7!}\right)$

Answer:

$$\binom{27}{18} - \left(10 + \frac{10!}{8!} + \frac{10!}{8!} + \frac{10!}{2!7!}\right)$$

$$\frac{a}{b} \quad x^y \quad \binom{n}{k}$$

- Q2) Consider the following programming segment. Your answer must rely on combinations structure. Answers that use sigma notation will not be accepted.

```

counter = 2000

for i = a+5 to 3n+1 do {

    counter = counter + 7

    for j = i+1 to 5n-1 do {

        counter = counter + 11

    }

}

// assume 3n+1 > a+5

```

- a)** Determine the value that the variable counter has after the segment is executed. Provide your answer as a function of n (i.e. formula which depends on n). Make sure to explain how/why the parts of the formula relate to counting.

Short Explanation: i loop runs from a+5 to 3n+1, that is $3n+1 - (a+5) + 1 = 3n - a - 3$ times. Considering i, j loops together note that from a + 5 to 5n - 1 there are $(5n - 1 - (a+5) + 1) = 5n - a - 5$ distinct numbers. The number of different pairs is $\binom{5n-a-5}{2}$. However, pairs of i,j made from 3n+2, 3n+3, ..., 5n - 1 are never executed. Since $5n - 1 - (3n+2) + 1 = 2n - 2$, there are such $\binom{2n-2}{2}$ such pairs.

Answer: $2000 + (3n - a - 3) \cdot 7 + \left[\binom{5n-a-5}{2} - \binom{2n-2}{2} \right] \cdot 11$

$$\frac{a}{b} \quad x^y \quad \binom{n}{k}$$

- Q3) Consider the following programming segment. Your answer must rely on combinations structure. Answers that use sigma notation will not be accepted.

```

a = 12

b = 34

c = 56

d = 78

for i = 5 to 4n do {
    a = a + 3

    d = b

    for j = i+1 to n do {
        b = b + 6
    }

    If (d = b) {
        c = c + 9
    }
}

// assume n>5

```

Determine the value that the variables a , b and c have after the segment is executed. Provide your answer as a function of n (i.e. formula which depends on n). Make sure to explain how/why the parts of the formula relate to counting.

Explanation: ***for counters a and b student should refer to Lab 2 ***

Counter c starts incrementing when b stops incrementing, that is for $i = n, n+1, n+2, \dots, 4n$, which is $(4n - n + 1) = 3n + 1$ increments.

Answer:

$$a = 12 + (4n - 4) \cdot 3$$

$$b = 34 + \binom{n-4}{2} \cdot 6$$

$$c = 56 + (3n + 1) \cdot 9$$

$$\frac{a}{b} \quad x^y \quad \binom{n}{k}$$

Q4) Use rules of inference to prove the argument is valid.

$$q \vee \neg c$$

$$(a \rightarrow b) \rightarrow (c \wedge d)$$

$$r \wedge \neg q$$

$$f \vee b \vee r$$

$$\therefore \neg b$$

STEPS		REASON
1	$r \wedge \neg q$	given
2	$\neg q$	CS 1
3	$q \vee \neg c$	given
4	$\neg c$	DS 2, 3
5	$\neg c \vee \neg d$	DA 4
6	$\neg (c \wedge d)$	De Morgan's 5
7	$(a \rightarrow b) \rightarrow (c \wedge d)$	Given
8	$\neg(a \rightarrow b)$	MT 6, 7
9	$\neg(\neg a \vee b)$	No Name 8
10	$\neg\neg a \wedge \neg b$	De Morgan's 9
11	$\therefore \neg b$	CS 11

This was a direct proof. Below we repeat this question using a proof by Contradiction.

$$\frac{a}{b} \quad x^y \quad \binom{n}{k}$$

Redone, using a proof by **Contradiction**

STEPS		REASON
1	b	Negating Conclusion, Attempting Proof by Contradiction
2	$(a \rightarrow b) \rightarrow (c \wedge d)$	given
3	$(\neg a \vee b) \rightarrow (c \wedge d)$	No Name 2
4	$b \vee \neg a$	DA 1
5	$c \wedge d$	MP 3, 4
6	c	CS 5
7	$r \wedge \neg q$	given
8	$\neg q$	CS 7
9	$q \vee \neg c$	given
10	$\neg c$	DS 8, 9
11	$\therefore \neg b$	Lines 6, 10 are in Contradiction

$$\frac{a}{b} \quad x^y \quad \binom{n}{k}$$

Q5) Use rules of inference to prove the argument is valid.

$$\neg d \rightarrow \neg c$$

$$(a \wedge p) \rightarrow b$$

$$\neg p \rightarrow c$$

$$z \wedge \neg b$$

$$\therefore \neg a \vee d$$

STEPS		REASON
1	$z \wedge \neg b$	given
2	$\neg b$	CS 1
3	$(a \wedge p) \rightarrow b$	given
4	$\neg(a \wedge p)$	MT 2, 3
5	$\neg a \vee \neg p$	De Morgan's 4
6	$a \rightarrow \neg p$	No Name 5
7	$\neg p \rightarrow c$	given
8	$a \rightarrow c$	LS 6, 7
9	$\neg d \rightarrow \neg c$	given
10	$c \rightarrow d$	Contrapositive 9
11	$a \rightarrow d$	LS 8, 10
12	$\therefore \neg a \vee d$	No Name 11

This was a direct proof. Below we repeat this question using a proof by Contradiction.

$$\frac{a}{b} \quad x^y \quad \binom{n}{k}$$

Redone, using a proof by **Contradiction**

STEPS		REASON
1	$\neg(\neg a \vee d)$	Negating Conclusion, Attempting Proof by Contradiction
2	$\neg\neg a \wedge \neg d$	De Morgan's 1
3	$a \wedge \neg d$	Double Negation 2
4	a	CS 3
5	$\neg d$	CS 3
6	$\neg d \rightarrow \neg c$	given
7	$\neg c$	MP 5, 6
8	$\neg p \rightarrow c$	given
9	p	MT 7, 8
10	$a \wedge p$	RC 4, 9
11	$(a \wedge p) \rightarrow b$	given
12	b	MP 10, 11
13	$z \wedge \neg b$	given
14	$\neg b$	CS 13
15	$\therefore \neg a \vee d$	Lines 12, 14 are in Contradiction